

The University of Lahore
Department of Computer Science & IT
Final Term Examination



Fall-2023

Course Title:	Data Structures & Algorithms	Course Code:	CS09204	Credit Hours:	3+1
Course Instructors:	Ms. Duaa-e-Nadeem, Ms. Nimra Shafiq, Ms. Kashmala Pervaiz, Mr. Muhammad Fiaz	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	50		
Date:	27-12-2023	Course Mentor:	Dr. Mehtab Afzal		
Student's Name:		Signature:			
Reg. No:		Section:			

Question # 1: (7 Marks)

Consider a hash table with a size of 11. The hash function is $h(k) = k \bmod 11$. Perform quadratic probing to insert the following keys into the hash table: 19, 22, 33, 44, 55 and 80. Show the final arrangement of keys in the hash table step by step.

Question # 2: (8 Marks)

Implement a variation of merge sort that sorts only the odd-indexed elements in the array while leaving the even-indexed elements in their original positions. Provide the code for the modified merge sort function.

Unsorted Array:

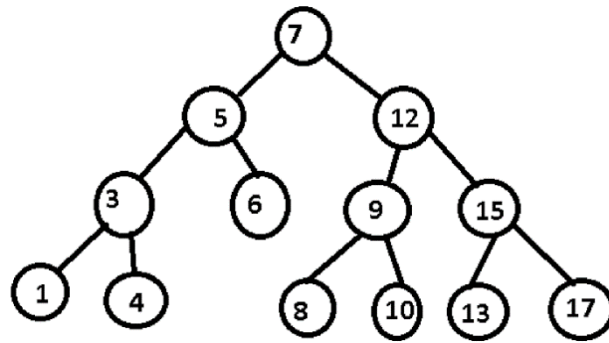
0	1	2	3	4	5	6	7	8
9	6	2	8	7	5	4	3	1

Sorted Array:

0	1	2	3	4	5	6	7	8
1	6	2	8	4	5	7	3	9

Question # 3: (10 marks)

- a) Consider the following Binary Search Tree. Remove value 5 and 12 from the tree one at a time and draw the new Binary Search Tree. Clearly describe each step.

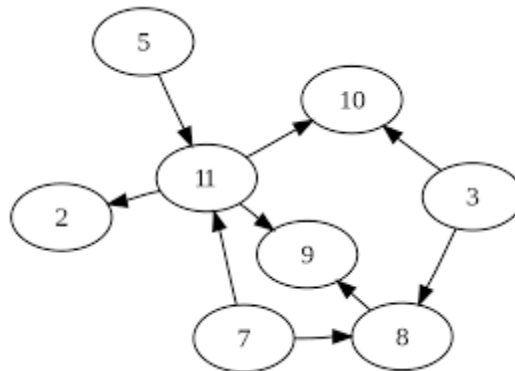


b) Given two binary trees, write a function in C++ to check if they are identical/same.

Question # 4:

(10 marks)

Consider the following unweighted directed graph. Draw its adjacency matrix and adjacency list.



Question # 5:

(15 marks)

Considering the following tree and tell if the tree is an AVL tree or not.

- Insert 12 first and then 16 into the tree. If the tree is not an AVL tree after insertion then show step by step process of converting a tree into an AVL tree.
- Show inorder traversal of the final tree which was constructed in part (a).
- Build a min heap from the tree constructed in part (a).

